1.4 Data types, data structures and algorithms	
How data is represented and stored within different structures. Different algorithms that can be applied to these structures	
1.4.1 Data Types	a) Represent positive integers in binary. b) Use of Sign and Magnitude and Two's Complement to represent negative numbers in binary. c) Addition and subtraction of binary integers. d) Represent positive integers in hexadecimal. e) Representation and normalisation of floating point numbers in binary. f) Floating point arithmetic, positive and negative numbers, addition and subtraction. g) Bitwise manipulation and masks: shifts, combining with AND, OR, and XOR. h) How character sets (ASCII and UNICODE) are used to represent text.
1.4.2 Data Structures	a) Arrays (of up to 3 dimensions). b) The following structures to store data: linked-list, graph (directed and undirected), stack, queue, tree, binary search tree, hash table. c) How to create, traverse, add data to and remove data from the data structures mentioned above. *(NB this can be* ***either*** *using arrays and procedural programming* ***or*** *an object-oriented approach).*
1.4.3 Boolean Algebra	a) Define problems using Boolean logic. b) Use the following rules to derive or simplify statements in Boolean algebra: De Morgan's Laws, distribution, association, commutation, double negation.